

B.Tech. Degree IV Semester Examination April 2014

ME 406 MANUFACTURING PROCESS
(2006 Scheme)

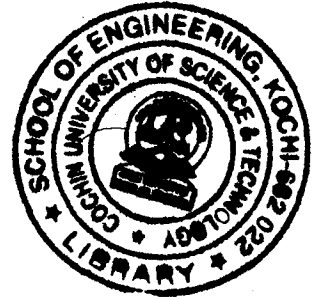
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) Define Casting.
(b) Discuss the need for providing pattern allowance.
(c) List out the desirable properties of moulding sand.
(d) Describe the basic principle of shell moulding.
(e) Distinguish between hot working and cold working.
(f) Explain the principle of rolling .
(g) Distinguish between fusion welding and non-fusion welding.
(h) Explain thermist welding.



PART B

(4 x 15 = 60)

- II. Explain in detail the different types of patterns with appropriate sketches.

OR

- III. What are the requirements of a good gating system? Also explain the significance of gating ratio.

- IV. Explain in detail the defects in casting. Also explain the methods to eliminate these defects.

OR

- V. Writes notes on :

- (a) Die casting
(b) Casting metallurgy

- VI. Describe the process of extrusion. Also distinguish between direct extrusion and indirect extrusion.

OR

- VII. Explain the different types of forging. Also explain the various defects in forging.

- VIII. Write notes on:

- (a) TIG welding
(b) Friction welding

OR

- IX. Distinguish between soldering and brazing.
